SUTEX BANK COLLEGE OF COMPUTER APPLICATIONS & SCIENCE, AMROLI
MSc IT PROGRAMME
MODEL QUESTION PAPER - 3

Course: FY MSc IT / BSc IT	**Year: 2026-27**
Semester: 1st	**Total Marks : 50**
Subject: 104 Fundamentals of Programming Using C-1	**Time: 2 Hours**

Q.1	**Do as directed: (Attempt any Ten)**	**[10]**
1	How many keywords are there in ANSI C standard?	
2	Which symbol is used for single-line comments in C?	
3	What is the value of 5 + 2 * 10 / 2 in C?	
4	In a flowchart, what is the purpose of the circle symbol?	
5	Which loop in C is guaranteed to execute at least once?	
6	True or False: High-level language programs are directly executed by CPU without translation.	
7	What will be printed by printf("%d", 10 != 10); in C?	
8	Which function is used in C to accept formatted input from the user?	
9	What is the range of values for an unsigned char (1 byte) in C?	
10	Which keyword is used to handle all unmatched cases in a switch statement?	

11	What will be the output of the following code snippet?
	`int i = 0;`
	`while(i < 3) {`
	`  printf("%d ", i);`
	`  i += 2;`
	`}`	

12	Give an example of shorthand assignment operator in C.	
Q.2 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	What is an Identifier in C? State all rules for naming identifiers/variables in C with valid and invalid examples. Explain Problem-Solving techniques: Simplification, Divide and Conquer, and Constraint Examination with examples. Explain the difference between break and continue statements in C loops with syntax, flow diagrams, and examples.	
Q.2 (B)	**Answer following question: (Attempt any One)**	**[04]**
(1)	Explain Bitwise operators in C (&, |, ^, ~, <<, >>) with simple examples.	

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OR (2)	Differentiate between Low-Level (Machine/Assembly) and High-Level Programming Languages.	
Q.3 (A)	**Answer following questions: (Attempt any Two)**	**[10]**
(1) (2) (3)	What is a Flowchart? Explain advantages and limitations of flowcharts in problem solving. Explain nested loops in C with an example program to print a right-angled triangle pattern of asterisks (*). What is Type Conversion? Explain automatic type promotion in expressions with suitable examples.	
Q.3 (B)	**Answer following question: (Attempt any One)**	**[04]**
(1) **OR** (2)	Explain the structure of a C program and explain the role of header files and #include preprocessor directive. What are runtime errors and logical errors? How can a programmer debug them?	
Q.4	**Answer following questions: (Attempt any Three)**	**[12]**
(1) (2) (3) (4)	Write a C program to swap two numbers using a third variable, and without using a third variable. Draw a flowchart and write an algorithm to find the largest of three numbers A, B, and C. Write a C program to generate the Fibonacci series up to N terms (0, 1, 1, 2, 3, 5, 8, ...). Write a C program to check whether a given integer number is an Armstrong number (e.g., 153 = 1^3 + 5^3 + 3^3).	